Identification of Cations

Exam code: 0620 & 0971

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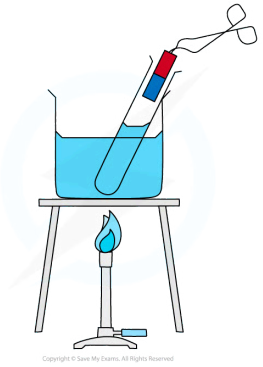
Updated on 15 October 2025

Identification of cations

Test for ammonium ions

- Ammonium ions, NH_4^+, can be identified by gently warming a solution containing the ions with sodium hydroxide solution
  - The sodium hydroxide solution is a source of hydroxide ions, OH^-, for the test
- This releases ammonia gas which turns damp red litmus paper blue

Testing for ammonium ions



Heating ammonium ions with sodium hydroxide solution releases ammonia gas which turns damp red litmus blue

- Metal cations in aqueous solution can be identified by the **colour** of the precipitate they form on addition of sodium hydroxide and ammonia
  - Most transition metals produce hydroxides with distinctive colours

Test for metal ions with sodium hydroxide solution

- If a small amount of sodium hydroxide solution is used, the resulting **metal hydroxide** normally precipitates out of solution
- If **excess** sodium hydroxide solution is used, some of the precipitates may re-dissolve

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Add a few drops of sodium hydroxide solution
- Record any colour changes or precipitates formed
- Add excess sodium hydroxide solution
- Record any colour changes or changes to precipitates

Test with sodium hydroxide summary

Metal Ion	Effect of aqueous NaOH	Effect of excess aqueous NaOH
aluminium, Al^{3+}	white precipitate	soluble in excess giving a colourless solution
ammonium, NH_4^+	ammonia produced on warming	-
calcium, Ca^{2+}	white precipitate	insoluble in excess
chromium(III), Cr^{3+}	green precipitate	soluble in excess
copper, Cu^{2+}	light blue precipitate	insoluble in excess
iron, Fe^{2+}	green precipitate	insoluble in excess, but the surface of the precipitate turns brown on standing
iron, Fe^{3+}	red-brown precipitate	insoluble in excess
zinc, Zn^{2+}	white precipitate	soluble in excess giving a colourless solution

Test for metal ions with ammonia solution

- If a small amount of ammonia solution is used, the resulting **metal hydroxide** normally precipitates out of solution
- If **excess** ammonia solution is used, some of the precipitates may re-dissolve
  - For this reason, just a few drops of ammonia solution are added at first and **very slowly**
- The ammonia test for the metal ion is:
  - Add a few drops of ammonia solution
  - Record any colour changes or precipitates formed
  - Add excess ammonia solution

Test with ammonia summary

Metal Ion	Addition of 2-3 drops of ammonia	Addition of excess ammonia
aluminium, Al^{3+}	white precipitate	insoluble in excess
calcium, Ca^{2+}	no precipitate or very slight white precipitate	no change
chromium(III), Cr^{3+}	green precipitate	insoluble in excess
copper, Cu^{2+}	light blue precipitate	soluble in excess giving a dark blue solution
iron, Fe^{2+}	green precipitate	insoluble in excess, but the surface of the precipitate turns brown on standing
iron, Fe^{3+}	red-brown precipitate	insoluble in excess
zinc, Zn^{2+}	white precipitate	soluble in excess giving a colourless solution

Analysing results

- The tables above contain the results for all **metal cations** included in the syllabus
- For example, zinc chloride:

$$ZnCl_2\,(aq) + 2NaOH\,(aq) \rightarrow Zn(OH)_2\,(s) + 2NaCl\,(aq)$$

- There are 3 metal ions that all form white precipitates:
  - Aluminium ions, Al^{3+}
  - Calcium ions, Ca^{2+}
  - Zinc ions, Zn^{2+}
- Calcium ions, Ca^{2+}, can be easily distinguished from Zn^{2+} and Al^{3+}
  - The white precipitate of calcium hydroxide **does not dissolve** in excess sodium hydroxide solution
  - The white precipitates of zinc hydroxide and aluminium hydroxide **dissolve** in excess sodium hydroxide solution
- Zinc ions, Zn^{2+}, can then be distinguished from Al^{3+} ions as
  - The white precipitate of zinc hydroxide **dissolves** in excess ammonia solution

Examiner Tips and Tricks

The ammonia or sodium hydroxide solution must be added very slowly. If it is added too quickly and the precipitate is soluble in excess, then you run the risk of missing the formation of the initial precipitate, which dissolves as quickly as it forms if excess solution is added.

Be sure to distinguish between the term "**colourless**" and "**clear**". A solution that loses its colour has become colourless. A clear solution is one that you can see through such as water. Solutions can be clear **and** have colour eg. dilute copper sulphate.

Flame tests for metal ions

- The **flame test** is used to identify the metal cations by the colour of the flame they produce
  - Ions from **different** metals produce **different colours**
- Dip the loop of an **unreactive** metal wire such as nichrome or platinum in concentrated acid and then hold it in the blue flame of a Bunsen burner until there is no colour change
  - This is an important step as the test will only work if there is just **one type** of ion present
    - Two or more ions means the colours will mix, making identification erroneous
    - This cleans the wire loop and avoids **contamination**
- A small sample of the compound is placed on an **unreactive** metal wire loop such as nichrome or platinum
- Dip the loop into the solid sample / solution and place it in the edge of the **blue** Bunsen flame
  - Avoid letting the wire get so hot that it glows red otherwise this can be confused with a flame colour

Diagram showing the technique for carrying out a flame test

Cation	Flame Colour
Li^+	Red
Na^+	Yellow
K^+	Lilac
Ca^{2+}	Orange-red
Ba^{2+}	Light-green
Cu^{2+}	Blue-green

Metal ions form distinctive coloured flames

Examiner Tips and Tricks

The sample needs to be heated strongly, so the Bunsen burner flame should be on a blue flame.

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